Real positive-definite counterexamples to extended q-permanent monotonicity

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Abstract

An explicit real symmetric positive-definite matrix of order five has an inversion-weighted q-permanent that decreases between $q = 14$ and $q = 15$. Its polynomial has nine nonzero permutation contributions. This refutes the real positive-definite monotonicity conjecture in da Fonseca's 2010 paper. A sparse parametric construction gives a stronger statement: for every prescribed real $q_0 > 1$, some real symmetric positive-definite matrix has a strictly negative q-permanent derivative at q_0. The dimension may depend on q_0. Lean 4 formalizations establish the finite example, the theorem for every prescribed parameter, and the correlation and integer-entry consequences. The result does not address real-symmetric monotonicity on $[-1, 1]$.

1 The statement and its scope

The ordinary permanent sums one entry product for each permutation. The q-permanent assigns an additional weight to each product. For an $n \times n$ matrix $A = (a_{ij})$, define

$$P_q(A) = \sum_{\sigma \in S_n} q^{\mathrm{inv}(\sigma)} \prod_{i=1}^{n} a_{i,\sigma(i)}, \qquad \mathrm{inv}(\sigma) = \#\{(i,j) : i < j,\ \sigma(i) > \sigma(j)\}.$$

The row and column order is fixed. In particular, simultaneous permutation of rows and columns need not preserve this polynomial. The convention is $0^0 = 1$.

Da Fonseca's 2010 paper explicitly adopts the convention that $A > 0$ denotes a real symmetric positive-definite matrix, before stating Conjecture 1 in Section 2 [1]. That conjecture asks for strict monotonicity on an interval extending from a left endpoint at or below -1 to $+\infty$. Its later restatement uses a left endpoint strictly below -1 [2, Conjecture 2]. Either formulation implies

$$P_{q_1}(A) < P_{q_2}(A) \qquad (-1 < q_1 < q_2) \tag{1}$$

for every admissible non-diagonal real symmetric positive-definite A. The counterexample contradicts (1) at two positive integer parameters, so the distinction between the two left-endpoint formulations is immaterial.

The Bapat–Lal monotonicity assertion is restricted to $[-1, 1]$. The October 2026 record of Kitamura's order-144 complex Hermitian counterexample explicitly leaves its real-symmetric case unsettled [4]. The present results concern $q > 1$.

The mechanism is the last square in the family formula below. At a chosen parameter, its derivative is negative. Intervening identity coordinates increase its power of q faster than the positive terms. This permits a strict decrease at any fixed $q > 1$ when the dimension is large enough.

Mitchell discusses a failure of the extended conjecture for positive semidefinite rank-one matrices at negative parameters [5]. That result does not supply the real positive-definite examples above one proved here. De Sá's earlier counterexamples concern incorrect tree expansion formulas [3]; the present proof uses a direct permutation sum and does not use those formulas.

2 A five-dimensional exact counterexample

Theorem 1. *The matrix*

$$A = \begin{pmatrix} 5 & 0 & 0 & 4 & 1 \\ 0 & 5 & 0 & -1 & 4 \\ 0 & 0 & 1 & 0 & 0 \\ 4 & -1 & 0 & 5 & 0 \\ 1 & 4 & 0 & 0 & 5 \end{pmatrix}$$

is real symmetric, non-diagonal, and positive definite. Its q-permanent is

$$\begin{aligned} P_q(A) &= 625 + 25q^3 + 800q^5 + 25q^7 + 256q^8 - 32q^9 + q^{10} \\ &= 625 + 25q^3 + 800q^5 + 25q^7 + q^8(q-16)^2. \end{aligned} \tag{2}$$

It satisfies

$$P_{14}(A) = 8\,968\,822\,249 > 7\,441\,960\,000 = P_{15}(A),$$

with exact difference $1\,526\,862\,249$*. Consequently the real monotonicity conjecture described above is false.*

Proof. For positivity alone, order the coordinates as $(1, 2, 4, 5, 3)$. The resulting matrix is the direct sum of [1] and

$$\begin{pmatrix} 5I_2 & R \\ R^T & 5I_2 \end{pmatrix}, \qquad R = \begin{pmatrix} 4 & 1 \\ -1 & 4 \end{pmatrix}, \qquad RR^T = 17I_2.$$

Its eigenvalues are 1 and $5 \pm \sqrt{17}$, with each of the latter occurring twice. All are positive. This coordinate permutation is used only to establish positive definiteness; the q-permanent is calculated in the original displayed order.

Every nonzero term fixes coordinate 3. The only possible nontrivial cycles among the remaining four coordinates are the four transpositions corresponding to nonzero off-diagonal entries, the two disjoint pairs of those transpositions, and the two oriented four-cycles. Together with the identity, these give exactly the nine terms in Table 1. Summing the table proves (2). Substitution of 14 and 15 yields the stated integer values, which contradict strict monotonicity. □

Permutation in one-line notation	Inversions	Entry product
$(1, 2, 3, 4, 5)$	0	625
$(1, 4, 3, 2, 5)$	3	25
$(1, 5, 3, 4, 2)$	5	400
$(4, 2, 3, 1, 5)$	5	400
$(4, 5, 3, 1, 2)$	8	256
$(4, 5, 3, 2, 1)$	9	-16
$(5, 2, 3, 4, 1)$	7	25
$(5, 4, 3, 1, 2)$	9	-16
$(5, 4, 3, 2, 1)$	10	1

Table 1: All nonzero terms for the fixed order in Theorem 1.

3 A sparse family and its complete polynomial

Let $m \geq 0$ be an integer and $n = m + 4$. The active coordinates are

$$i = 1, \qquad j = 2, \qquad k = m + 3, \qquad \ell = m + 4.$$

For real $a \geq 0$ and $d > \sqrt{a^2+1}$, define $A_m(a,d)$ to be the identity on the m remaining coordinates, with zero couplings to them, and with the following principal submatrix on the active coordinates in their increasing order:

$$\begin{pmatrix} d & 0 & a & 1 \\ 0 & d & -1 & a \\ a & -1 & d & 0 \\ 1 & a & 0 & d \end{pmatrix}. \tag{3}$$

It is positive definite, since the active block has eigenvalues $d \pm \sqrt{a^2+1}$, each twice. The intervening identity coordinates are part of the fixed ordering and must not be deleted when counting inversions.

Lemma 2. *For every $m \geq 0$,*

$$P_q(A_m(a,d)) = d^4 + d^2 q^{2m+1}(1 + 2a^2q^2 + q^4) + q^{4m+4}(a^2 - q)^2. \tag{4}$$

Proof. The isolated coordinates are fixed in every nonzero term. The nine possible active permutations are listed below using cycle notation; their inversion counts are computed in the full order $1, \ldots, m+4$.

Active permutation	Inversions	Entry product
id	0	d^4
$(j\ k)$	$2m+1$	d^2
$(i\ k)$, $(j\ \ell)$	$2m+3$	a^2d^2 each
$(i\ \ell)$	$2m+5$	d^2
$(i\ k)(j\ \ell)$	$4m+4$	a^4
$(i\ k\ j\ \ell)$, $(i\ \ell\ j\ k)$	$4m+5$	$-a^2$ each
$(i\ \ell)(j\ k)$	$4m+6$	1

For a transposition of positions $r < s$, the inversion count is $2(s-r)-1$. For each of the last three types, the four active coordinates contribute respectively 4, 5, and 6 inversions. Each intervening fixed coordinate contributes four additional inversions. These facts prove the stated counts. Summing gives

$$d^4 + d^2q^{2m+1} + 2a^2d^2q^{2m+3} + d^2q^{2m+5} + a^4q^{4m+4} - 2a^2q^{4m+5} + q^{4m+6},$$

which is (4). □

4 Failure at every prescribed parameter above one

Theorem 3. *For every real $r > 1$, there are an integer $n \geq 5$ and a real symmetric positive-definite matrix A of order n such that*

$$\frac{d}{dq} P_q(A)\Big|_{q=r} < 0.$$

Thus no interval $[1, 1+\delta]$ with $\delta > 0$ can enjoy q-permanent monotonicity uniformly over all real symmetric positive-definite matrices of all orders.

Proof. Fix $r > 1$. Choose an integer $m \geq 1$ sufficiently large that

$$r^{2m+5} > (4m+4)(2m+5)\left(\frac{5r}{4} + 2\right)\left(1 + \frac{5}{2}r^3 + r^4\right). \tag{5}$$

Such an integer exists: the left side grows exponentially in m, whereas the right side is a quadratic polynomial in m times a constant depending only on r.

Set

$$K = 4m + 4, \qquad H = 2m + 1, \qquad t = r\left(1 + \frac{1}{K}\right), \qquad a = \sqrt{t}, \qquad d = \sqrt{t+2}.$$

These parameters are fixed after this choice; differentiation below is with respect to q only. Then $d > \sqrt{a^2+1}$, so $A = A_m(a,d)$ is positive definite. By Lemma 2,

$$P_q(A) = d^4 + d^2 q^H (1 + 2tq^2 + q^4) + q^K (t-q)^2.$$

The derivative of the last summand at $q = r$ is exactly

$$\begin{aligned} Kr^{K-1}(t-r)^2 - 2r^K(t-r) &= Kr^{K-1}\frac{r^2}{K^2} - 2r^K\frac{r}{K} \\ &= -\frac{r^{K+1}}{K}. \end{aligned}$$

The derivative of the middle summand is

$$(t+2)r^{H-1}(H + 2t(H+2)r^2 + (H+4)r^4).$$

Since $K \geq 4$, the bound $t \leq 5r/4$ holds. Also $H, H+2 \leq H+4 = 2m+5$. Consequently this positive derivative is at most

$$\left(\frac{5r}{4} + 2\right) r^{H-1}(2m+5)\left(1 + \frac{5}{2}r^3 + r^4\right).$$

Inequality (5), multiplied by r^{H-1}/K, says that this upper bound is strictly smaller than r^{K+1}/K, because $K - H + 2 = 2m + 5$. The total derivative is therefore negative.

Finally, given $\delta > 0$, apply the theorem with $r = 1 + \delta/2$. A negative derivative of a polynomial gives a strictly decreasing pair of sufficiently nearby parameters contained in $[1, 1+\delta]$. □

Corollary 4. *The matrices in Theorems 1 and 3 can be replaced by real positive-definite correlation matrices, with the same strict monotonicity failures. At every prescribed $r > 1$, a counterexample can additionally be chosen with no zero entries.*

Proof. If $D = \operatorname{diag}(s_1, \ldots, s_n)$ with all $s_i > 0$, then term by term

$$P_q(DAD) = \left(\prod_{i=1}^{n} s_i^2\right) P_q(A).$$

Choose $s_i = a_{ii}^{-1/2}$ to normalize the diagonal. Positive definiteness is preserved, and the multiplicative factor is positive and independent of q.

For the final assertion, first replace a matrix from Theorem 3 by $A + \varepsilon \mathbf{1}\mathbf{1}^T$. This remains positive definite for every $\varepsilon > 0$. For sufficiently small positive ε, all entries are nonzero and the strict negative derivative persists by continuity of the finite polynomial in the entries. Then normalize the diagonal as above. □

Corollary 5. *For every real $r > 1$, a real symmetric positive-definite matrix with integer entries can be chosen such that $P_r'(A) < 0$.*

Proof. Positive definiteness and the strict inequality $P_r'(A) < 0$ are open conditions in the real vector space of symmetric matrices. Approximate the matrix of Theorem 3 by a sufficiently close rational symmetric matrix B preserving both conditions. Multiplication by a positive common denominator c gives an integer matrix, and $P_r'(cB) = c^n P_r'(B) < 0$. □

5 Formal verification and independent checks

The Lean development is in `lean/MathAndTheory/Mathematics/QPermanent/`. Its namespace is `MathAndTheory.QPermanent`. The following declarations give the main correspondence.

Paper statement	Lean declaration
Theorem 1, positivity	`counterexample_posDef`
Theorem 1, polynomial	`counterexample_polynomial`
Source conjecture is false	`not_daFonseca_extended_conjecture`
Lemma 2	`qPermanent_family_formula`
Theorem 3	`exists_posDef_negative_qPermanent_derivative_ge_five`
Correlation consequence	`exists_correlation_counterexample`
Dense correlation consequence	`exists_dense_correlation_negative_qPermanent_derivative`
Integer-entry consequence	`exists_integer_posDef_negative_qPermanent_derivative`

The formal definition sums over every permutation and counts ordinary inversions in the displayed coordinate order. The proof establishes the contribution of every intervening coordinate. Positive definiteness is proved by quadratic forms. The integer-entry proof uses rational approximations to the explicit family parameters, followed by denominator clearing. This gives the same existence conclusion as Corollary 5.

The independent program reconstructs the five-dimensional polynomial by all 120 permutations. It checks the positive rational LDL^T pivots $5, 5, 1, 8/5, 8/5$, both endpoint values, and $P'_{15}(A) = -1\,563\,030\,000$. For $m = 0, 1, 2, 3, 4$, it independently enumerates all permutations of the corresponding family matrix. It also checks the growth condition and negative derivative with exact rational arithmetic at $r = 101/100, 11/10, 3/2, 2, 15$. The respective choices of m are $925, 69, 12, 6, 1$. These finite checks do not prove the theorem for every r. Theorem 3 and its formalization supply that proof.

The repository pins Lean 4.35.0 and Mathlib commit `56a005855ea17e0b0958f73ec801ae140b0a1679`. See `lean/docs/q-permanent.md` for statement alignment and `verification/q-permanent/` for commands, software versions, and recorded results. The Challenge states the permutation definition directly. It has no imports from this proof development. Comparator checks the corresponding Solution against those statements.

6 Scope and provenance

The construction gives an upper bound on a witness dimension, not a minimal dimension. The integer and correlation conclusions are separate; no integer correlation matrix is asserted. The results do not settle the real Bapat–Lal assertion on $[-1, 1]$.

The literature audit compared the original conjecture, its 2018 restatement, Mitchell's discussion, de Sá's correction, and the current public record of the complex counterexample. It found no earlier proof of the specific real positive-definite results stated here. This bounded search does not establish priority.

Dev D. Goyal directed the AI research and is responsible for publication. ChatGPT (OpenAI), using the GPT-6 model with Pro reasoning, produced the supplied construction, proofs, and Lean development. Codex audited the source statements, checked the mathematics, adapted the formalization, performed independent computations, and edited this manuscript. The repository records the later AI referee review. Machine verification and AI review do not constitute human expert peer review. This manuscript has not been peer reviewed.

References

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[4] Kenta Kitamura. Lean 4 disproof of q-permanent monotonicity (order 144), 2026. URL `https://github.com/KitaKen1/bapat-lal-q-permanent-lean/tree/4200da4fc1a132d69c23fb877795b5b72089544b`. Public formalization; immutable source revision.

[5] L. Mitchell. A note on Bapat's q-permanent conjecture. *Operators and Matrices*, 14(4): 915–919, 2020. doi: 10.7153/oam-2020-14-56.